

Department of Geology and Geophysics

Subject name: Electrical methods Lab (0-0-3) -2 Credit

Pre-requisite: None

Lab measurement of resistivity and dielectric constant, model experiments on DC resistivity, geophysical construction of 2, 3 and multi-layer curves, computer assisted computation of apparent resistivity, Interpretation of two layer, three layer and multi-layer curves using auxiliary point chart and master curves, Interpretation of field curve using direct and indirect methods by computer, Testing of equivalence and suppression in electrical. Analytical computation of profiling, pseudo depth sections. Forward modeling in Self-potential method. 2-D and 3-D resistivity imaging approaches.

Text Books

1. Applied Geophysics, Telford et al, 1990
2. Geosounding Principles I, Kefoed O, 1979.
3. Direct Current Electric sounding, Bhattacharya PK. And Patra, HP, 1968.

Reference books

1. Potential Theory and Applied Geophysics, K.K. Roy
2. An Introduction to Geophysical Exploration, P. Keary

S.No	Topic	Number of lectures
1	Measurement of resistance using DDR1 resistivity meter	1
2	Measurement of resistance using DDR3 resistivity meter	1
3	Measurement of resistance using Aqua meter	1
4	Potential distribution over a graphite coated sheet with inhomogeneity	1
5	Computation of SP anomaly over sphere, cylinder and sheet	1
6	Interpretation of SP anomaly over sphere, cylinder and sheet	1
7	Computation of apparent resistivity over 2-layer model	1
8	Interpretation of resistivity data over 2 layer model	1
9	Computation of apparent resistivity over multilayered earth using digital linear filtering approach	3
10	Interpretation of resistivity data over multi-layered earth	2
11	2-D and 3-D resistivity measurement	2
	Total	15